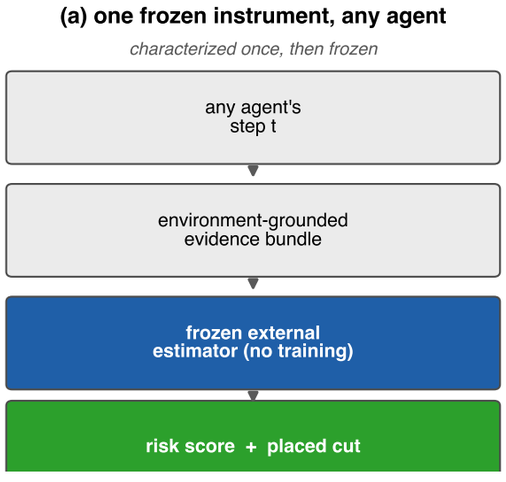
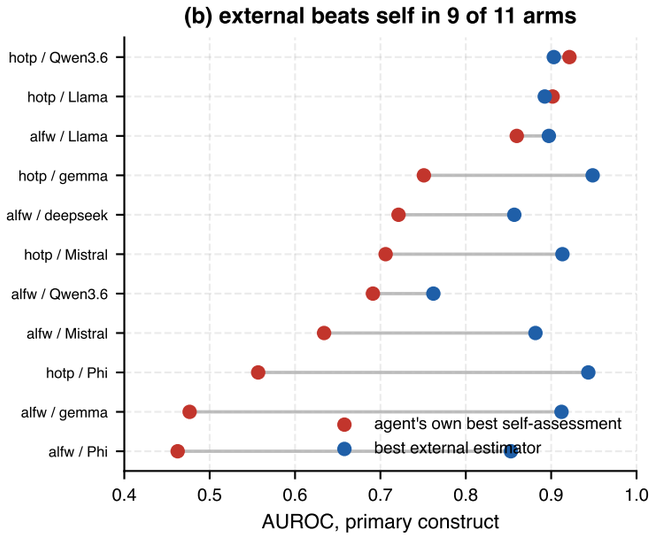
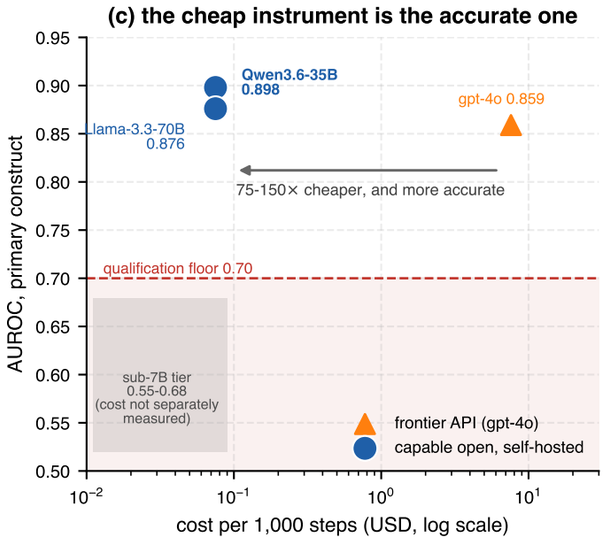
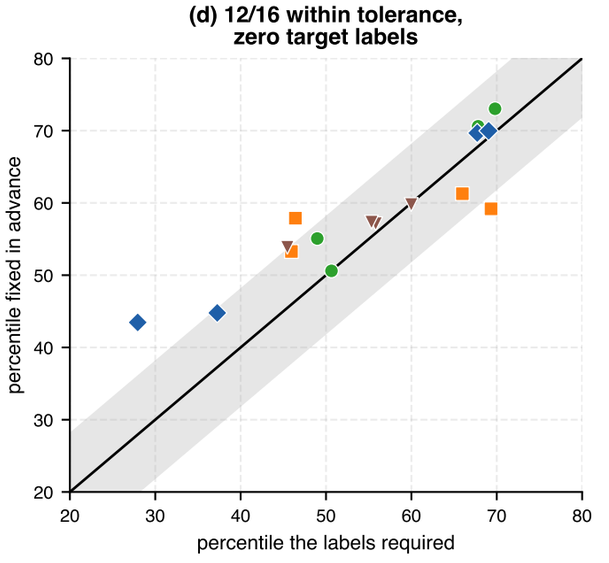

(a) one frozen instrument, any agent
characterized once, then frozen
any agent's step t
environment-grounded evidence bundle
frozen external estimator (no training)
risk score + placed cut


(b) external beats self in 9 of 11 arms
hotp / Qwen3.6
hotp / Llama
alfw / Llama
hotp / gemma
alfw / deepseek
hotp / Mistral
alfw / Qwen3.6
alfw / Mistral
hotp / Phi
alfw / gemma
alfw / Phi
agent's own best self-assessment
best external estimator
0.4
0.5
0.6
0.7
0.8
0.9
1.0
AUROC, primary construct


(c) the cheap instrument is the accurate one
Qwen3.6-35B
0.898
Llama-3.3-70B
0.876
gpt-4o 0.859
75-150× cheaper, and more accurate
qualification floor 0.70
sub-7B tier
0.55-0.68
(cost not separately measured)
frontier API (gpt-4o)
capable open, self-hosted
AUROC, primary construct
cost per 1,000 steps (USD, log scale)


(d) 12/16 within tolerance,
zero target labels
percentile fixed in advance
percentile the labels required